

KITTLA, Finland

WMO: 027200

Lat: 67.696N

Lon: 24.858E

Elev: 644

StdP: 14.36

Time Zone: 2.00 (EUE)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.6	-19.9	-31.3	1.0	-25.0	-25.2	1.4	-19.8	22.9	24.6	20.5	23.1	1.9	250	0.722

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.0	77.0	61.5	73.1	59.4	69.3	58.0	63.7	72.7	61.4	69.4	59.4	66.4	7.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.5	80.6	66.5	58.1	73.8	63.7	56.3	69.1	61.9	29.3	72.6	27.7	69.2	26.3	66.6	70.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	19.2	16.7	14.5	DB	-32.5	80.6	5.1	3.7	-36.2	83.2	-39.2	85.3	-42.1	87.4	-45.8	90.1
(5)				WB	-32.7	66.2	5.1	2.8	-36.4	68.2	-39.4	69.8	-42.2	71.4	-45.9	73.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	32.6	9.6	9.9	19.0	31.3	42.6	51.8	58.2	53.8	45.2	31.9	20.6	16.0
	DBStd	19.39	13.61	14.43	10.74	7.13	8.04	6.76	6.30	6.00	6.28	9.05	11.03	12.35
	HDD50	6914	1253	1122	962	560	258	57	7	29	167	561	883	1055
	HDD65	11838	1718	1542	1427	1010	694	398	225	352	593	1026	1333	1520
	CDD50	570	0	0	0	0	28	111	261	146	24	0	0	0
	CDD65	20	0	0	0	0	0	3	14	3	0	0	0	0
	CDH74	241	0	0	0	0	11	36	163	31	0	0	0	0
	CDH80	28	0	0	0	0	0	4	20	4	0	0	0	0
Wind	WSAvg	6.1	5.3	5.6	6.8	6.7	7.0	7.0	6.2	5.7	6.1	6.0	5.3	5.8
Precipitation	PrecAvg	23.3	1.6	1.3	1.1	1.3	1.5	2.5	3.3	2.8	2.2	1.9	2.0	1.7
	PrecMax	29.8	3.1	3.0	2.4	3.0	3.8	4.2	5.6	5.7	4.4	3.7	3.8	3.0
	PrecMin	17.4	0.4	0.5	0.4	0.4	0.3	0.4	1.3	1.3	0.6	0.5	0.6	0.4
	PrecStd	3.6	0.6	0.6	0.6	0.6	0.8	0.9	1.2	1.1	1.1	0.8	0.8	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.1	39.4	44.4	57.0	75.2	79.1	82.7	78.4	64.2	51.9	42.7	39.5
		MCWB	33.0	35.9	36.8	45.4	56.8	63.4	63.7	63.8	54.7	48.6	42.4	36.6
	2%	DB	31.9	34.0	39.2	49.7	68.2	73.0	78.7	73.2	60.8	48.5	37.8	35.5
		MCWB	31.2	31.8	34.1	40.9	54.3	58.0	61.9	60.5	53.7	46.1	36.9	34.3
	5%	DB	29.8	31.6	35.9	46.0	62.1	68.0	74.9	69.1	57.8	46.4	35.3	33.4
		MCWB	29.3	30.7	32.3	38.5	50.1	55.5	60.8	59.0	52.2	44.7	34.8	32.6
	10%	DB	26.9	28.6	33.4	42.1	56.8	64.2	70.9	64.8	55.3	43.2	33.4	31.7
		MCWB	26.5	27.9	30.8	36.2	47.0	53.3	59.1	57.3	51.3	41.9	33.0	31.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.3	36.3	37.2	45.5	57.8	65.0	67.1	66.2	57.6	49.5	42.0	36.8
		MCDB	33.8	39.5	42.7	55.6	71.4	76.2	76.5	75.2	61.3	50.7	42.1	39.0
	2%	WB	31.3	32.4	34.7	40.9	54.6	60.5	64.3	62.8	55.1	46.5	36.9	34.3
		MCDB	31.7	33.6	38.2	47.8	66.8	67.9	73.9	70.0	58.6	48.0	37.5	35.0
	5%	WB	28.8	30.5	32.7	38.7	50.9	57.2	62.3	60.2	53.3	44.3	34.2	32.4
		MCDB	29.3	31.5	35.0	44.8	60.2	64.8	71.3	66.3	56.7	45.9	34.6	32.8
	10%	WB	26.4	27.9	31.1	36.7	47.8	54.9	60.4	58.2	51.7	42.0	32.6	30.7
		MCDB	26.9	28.5	33.0	41.2	55.4	62.0	68.3	63.4	54.8	43.2	32.8	31.3
Mean Daily Temperature Range	5% DB	MDBR	14.1	14.6	17.4	14.9	15.4	14.8	16.0	16.5	13.1	10.0	10.8	13.5
		MCDBR	13.1	11.0	15.5	19.2	25.2	21.8	23.5	23.5	17.5	11.0	9.0	9.8
		MCWBR	12.9	9.8	11.7	12.1	13.0	10.3	10.6	12.5	11.1	8.8	8.7	9.3
	5% WB	MCDBR	13.2	10.9	13.1	17.7	22.7	17.7	19.6	20.0	13.6	9.9	8.4	9.9
		MCWBR	13.0	10.0	10.4	11.5	12.5	9.8	10.1	11.6	9.4	8.4	8.3	9.6
Clear-Sky Solar Irradiance	taub	0.189	0.251	0.286	0.317	0.335	0.338	0.348	0.333	0.304	0.271	0.349	N/A	
	taud	1.925	2.094	2.182	2.216	2.429	2.485	2.468	2.493	2.550	2.336	2.740	N/A	
	Ebn at Noon	41	176	238	263	273	275	267	260	241	174	54	0	
	Edh at Noon	3	16	25	33	30	29	29	25	18	12	4	0	
All-Sky Solar Radiation	RadAvg	14	165	639	1241	1530	1590	1482	1086	567	190	32	0	
	RadStd	1	23	49	74	149	119	146	87	62	35	4	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page